

# THEORETICAL AND EXPERIMENTAL WORKS ON FAST REACTOR PHYSICS IN JAPAN

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## 1. Introduction

Basic research on fast reactor physics was started at the Japan Atomic Energy Research Institute (JAERI) in 1958; an exponential experiment on a natural uranium system was conducted in 1961. Researchers have recognized the utility of the pulsed neutron method and a series of nano-second order diffusion experiments has been made since 1962. Theoretical studies were started on the derivation of group constants in the Japanese Nuclear Code Committee. The code ESELEM, able to treat up to 1500 fine groups, was developed to obtain accurate spectra and effective cross-sections.

A fast breeder reactor development programme was prepared by JAERI in 1963. Construction of a horizontal, split-table type critical facility was started in 1964, which makes it possible to study the reactor physics of large fast cores for an extended period. The nuclear characteristics of zoned systems were studied, since these systems are necessary for the investigation of large fast cores by limited amounts of fuel. Various methods for obtaining the effective cross-sections, for solving the diffusion equation, and for evaluating the heterogeneity effect were investigated.

In the Atomic Energy Society of Japan, a fast reactor committee was organized in 1964. Theoretical researches on fast reactor physics and feasibility studies of large fast reactors have been carried out not only in JAERI but also in other universities and in industry. The fast reactor safety analysis group of the Japanese Nuclear Code Committee has engaged in developing several codes for the evaluation of fast reactor safety problems. Methods for measuring the various power coefficients of reactivity have been investigated. Other problems of interest are the improvement of experimental techniques,

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especially for spectrum measurements, and the development of noise techniques for the investigation of space dependent kinetics. These techniques will be incorporated into the fast critical experiment.

## 2. Group Constants

A multi-group approximation is always used for predicting the static characteristics of a reactor. However, there are doubts whether multi-group cross-sections can predict all the characteristics accurately, and whether the adjoint equation derived can be used as the importance equation. If multi-group cross-sections are defined properly, they should give the same expectation values of measurements as those obtained without group approximations. From perturbation theory, it was concluded that all the group constants should be averaged with the angular flux as a weight.<sup>(1)</sup> When reactivity effects caused by some perturbation are required, the operator representing this perturbation should be defined from the difference between the group constants of the perturbed system and those of the unperturbed system.

For the calculations of a neutron spectrum and effective cross-sections, the code ESELEM has been developed.<sup>(2)</sup> In this type of code, the fission spectrum and some cross-sections like inelastic cross-sections are usually approximated as constants in each coarse group. These approximations cause deep dips, in the calculated spectrum, which should not exist. The dips make smaller the elastic and inelastic removal cross-sections, and then introduce an error in the calculation of reactivity effects. These drawbacks are eliminated in the code ESELEM. For the sake of consistency and accuracy, all the nuclear data except inelastic matrices are provided for each fine group. The spectrum is calculated for the fundamental mode solution and its angular dependence is treated by either  $P_1$  or  $B_1$  approximation. In the degradation integrals, the scattering properties of all the elements are rigorously taken into account, because they influence the spectrum appreciably.

The lower energy limit can be down to 3 eV in the code. The data, however, are provided at present for the range from 60 eV to 10 MeV. A standard spectrum for obtaining fine group data is selected, so that the averaging process becomes simple. The group width is so small that the error introduced by the standard spectrum is negligible. For example, if the width is chosen as 0.01 in the lethargy unit, the error may not exceed 1% for almost all the nuclear data of light and medium elements. In the code ESELEM, the width is chosen as 0.00995 and the whole range from 60 eV to 10 MeV is divided into 1200 fine groups. Except for the elastic scattering the nuclear data are averaged by  $1/E$ -spectrum in each fine group. For the calculation of elastic transfer cross-sections, a constant spectrum is used.

Data have been prepared for eight elements,  $\text{Pu}^{239}$ ,  $\text{U}^{235}$ ,  $\text{U}^{238}$ ,  $\text{C}^{12}$ ,  $\text{O}^{16}$ ,  $\text{Na}^{23}$ , Al, Fe. The data for other elements will be provided in the near future.

Most of the nuclear data were taken from KFK-120, BNL-325, and BNL-400. The differential cross-sections of elastic scattering were given in the form of a series of Legendre polynomials either in the centre-of-mass or laboratory system. A spectrum obtained by this code is shown in Fig. 1. The dips indicated by arrows in the figure correspond to the predominant resonances of  $O^{16}$ ,  $Na^{23}$ , and  $Fe^{56}$ , respectively.

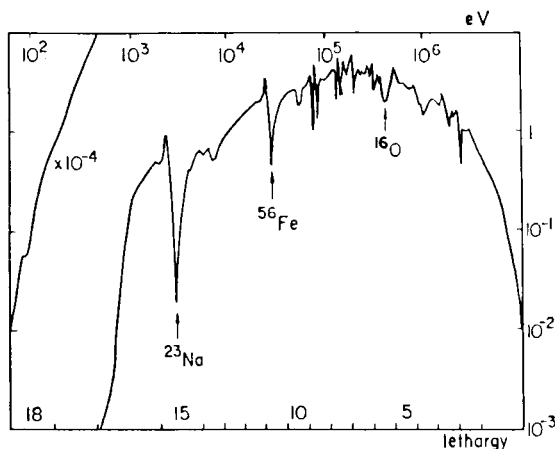


FIG. 1. Neutron spectrum (full sodium).

To define the effective fine group cross-sections of medium elements in the several tens of keV region, four methods are examined, namely, the methods, using the exact spectrum, an approximate solution of the slowing down,  $1/(E\Sigma_t)$  and  $1/E$ .  $1/(E\Sigma_t)$ -spectrum was found to give practically accurate values. A sub-group of the Japanese Nuclear Data Committee is now planning to prepare a standard cross-section set. The processing codes are now in development and the ENDF format is adopted as the standard format of nuclear data.

### 3. Numerical Methods

For criticality calculations, many different codes based on either  $S_N$  or diffusion have been developed.<sup>(3-6)</sup>

In the criticality calculation of various systems, having the same dimensions and geometries, it may be possible to reduce the machine time, if a higher order perturbation method is employed. To obtain higher mode solutions, the following procedure was used:<sup>(7)</sup> The initial source distribution of the  $i$ th mode was given by

$$S_i = \prod_{j=1}^{i-1} (L - \lambda_j F) \phi_i$$

where  $L$  and  $F$  denote the operators, representing the loss and creation of

neutrons, respectively, and  $\lambda_j$  denotes the  $j$ th eigenvalue. The trial function  $\phi_i$  is taken arbitrarily, but it is not the solution of  $\nabla^2 \phi_i = 0$  and  $\phi_i \neq \phi_j$ . The first iteration  $\phi^{(1)}$  is found by  $L^{-1}S_i$ . Then calculations are carried out in the same manner as for the fundamental mode solution. By this formulation the orthogonality is ensured. However, during the computation process, the orthogonality does not hold and, hence, the iteration  $\phi_i^{(n)}$  should be corrected. The method has been applied to one-dimensional calculations.

The general solution of the equation  $\nabla^2 \phi - B^2 \phi = -\rho$  can be represented by three terms. The first term is the special solution for the above equation,

$$\phi_\rho = \frac{1}{2\pi} \int_V K_0(BR) \rho(\vec{\gamma}_0) d\vec{\gamma}_0.$$

The second term is the solution of the homogeneous equation for the surface source distribution

$$\phi_\mu = \frac{1}{2\pi} \int_L K_0(BR) \mu(\sigma) d\sigma.$$

The third term is the solution for the double layer source distribution, that is

$$\phi_\delta = \frac{1}{2\pi} \int_L \frac{\partial}{\partial \nu} K_0(BR) \delta(\sigma) d\sigma.$$

$K_0$  is the modified Bessel function,  $L$  represents the contour of the region  $V$ , and  $\nu$  denotes the normal at  $\sigma$ .

The properties of these terms were studied:<sup>(8)</sup>  $\phi_\delta$  and the first normal derivative of  $\phi_\mu$  are continuous except on  $L$ . The limiting values from the left and right of  $L$  are different from each other. When the continuity of flux and current and the outermost boundary condition are imposed on the solution, an integral equation for  $\mu(\sigma)$  and  $\delta(\sigma)$  is derived. Moreover, if  $\mu$  is equal to zero on the outer boundary and  $\delta$  to zero on the inner boundary, the integral equation becomes the second kind equation and can be easily solved. For a one-group, two-region, cylindrical system a simultaneous equation was obtained, instead of the integral equation. The method was found to be useful. For the two-dimensional rectangular system, programming is now in progress.

For the treatment of the heterogeneity in a slab lattice, a new method, based on the superposition principle, has been developed.<sup>(9)</sup> The flux is expressed by a linear combination of eigen-solutions. The expansion coefficients are evaluated by solving the singular integral equation. The solution was found by two methods. In the first method, the solution of the homogeneous equation was expanded in a power series of a parameter which indicates the interference of the regions. In the second method, the original equation was converted into the Fredholm's equation of second kind, which was then solved. Three codes, able to treat up to five regions, have been

developed. A typical result is shown in Fig. 2. The accuracy of various approximations was examined for a two-region lattice. Although the present treatment requires more machine time than the conventional ones, the treatment is useful for finding the correction for the lattice.

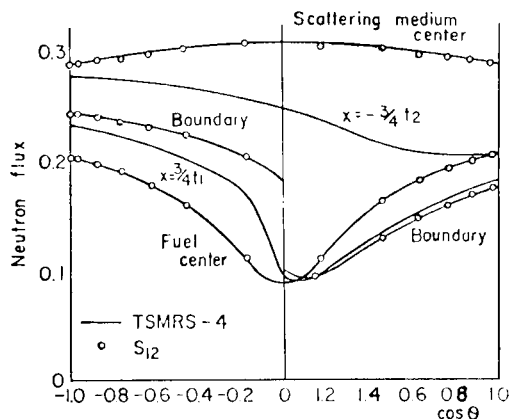


FIG. 2. Angular neutron flux.

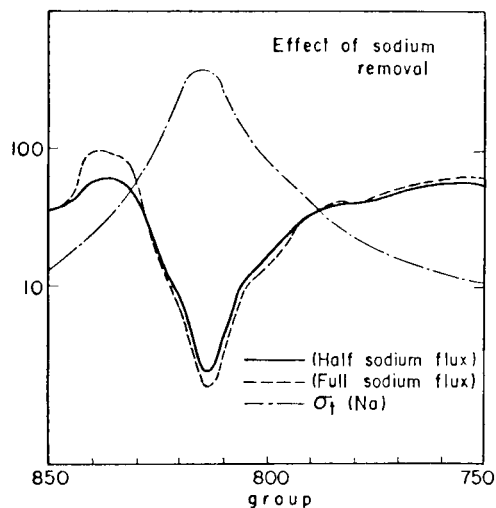


FIG. 3. Neutron spectra near 2.98 keV resonance of Na.

#### 4. Nuclear Characteristics

The characteristics of zoned systems, consisting of more than three zones, have been studied.<sup>(10)</sup> The reactivity change owing to sodium voids was evaluated for both the central and annular regions, because the leakage

TABLE 1. SODIUM VOID EFFECT ( $\Delta k/k\%$ ), FULL-HALF SODIUM

| Region                | Test zone          |                    |                    |                     | Driver zone        |                    |                    |
|-----------------------|--------------------|--------------------|--------------------|---------------------|--------------------|--------------------|--------------------|
|                       | I                  | II                 | III                | Sum<br>(I, II, III) | IV                 | V                  | Sum<br>(IV, V)     |
| Reactivity<br>Leakage | -0.0123            | -0.0818            | -0.0888            | -0.1829             | -0.3422            | -0.6254            | -0.9676            |
| Absorption            | +0.0047            | +0.0306            | +0.0323            | +0.0676             | +0.0580            | +0.0471            | +0.1051            |
| Scattering            | +0.2707<br>-0.2554 | +1.8201<br>-1.7344 | +2.0648<br>-2.0116 | +4.1556<br>-4.0014  | +3.5786<br>-3.5684 | +2.5410<br>-2.4910 | +6.1196<br>-6.0594 |
| Fission               | -0.0005            | -0.0027            | -0.0015            | -0.0047             | -0.0230            | -0.0221            | -0.0451            |
| Total                 | +0.0072            | +0.0318            | -0.0048            | +0.0342             | -0.2970            | -0.5504            | -0.8474            |

TABLE 2. REACTIVITY CHANGE DUE TO DOPPLER EFFECT  $\Delta\rho \times 10^5$  (1000-750°K)

| Group |                    | Core             |                   |         | Driver           |                  |         |                   | Blanket          |         |
|-------|--------------------|------------------|-------------------|---------|------------------|------------------|---------|-------------------|------------------|---------|
|       |                    | U <sup>238</sup> | Pu <sup>239</sup> |         | U <sup>238</sup> | U <sup>235</sup> |         | Pu <sup>239</sup> | U <sup>238</sup> |         |
| No.   | E <sub>L</sub> keV | Capture          | Absorption        | Fission | Capture          | Absorption       | Fission | Absorption        | Fission          | Capture |
| 3     | 110.0              | -0.1652          | -0.05000          | +0.1270 | -0.1023          | -0.1260          | +0.2758 | -0.05343          | +0.1463          | -0.6446 |
| 4     | 9.0                | -3.847           | -0.4452           | +1.003  | -2.597           | -1.324           | +2.527  | -0.4743           | +1.012           | -16.48  |
| 5     | 2.0                | -4.790           | -0.5972           | +1.363  | -4.188           | -2.196           | +3.815  | -0.6752           | +1.266           | -14.81  |
| 6     | 0.5                | -5.575           | -1.696            | +3.512  | -4.644           | -4.121           | +7.046  | -1.602            | +2.683           | -8.310  |
| Total |                    | -14.38           | -2.788            | +6.004  | -11.53           | -7.766           | +13.66  | -2.805            | +5.107           | -40.24  |
|       |                    | -14.38           | +3.216            |         | -11.53           | +5.898           |         | +2.302            |                  | -40.24  |

TABLE 3. CHARACTERISTICS OF Pu-U AND Pu-Th FAST REACTORS

| System                                             | (vol. %)<br>Fuel/SS/Na/void | Pu/U<br>or<br>Pu/Th | Breeding ratio    |                   | (γ)<br>Doubling**<br>time | (%)<br>(Δk) <sub>void</sub> † | (×10 <sup>-2</sup> )<br>β <sub>eff</sub> | (μsec)<br>l |
|----------------------------------------------------|-----------------------------|---------------------|-------------------|-------------------|---------------------------|-------------------------------|------------------------------------------|-------------|
|                                                    |                             |                     | Internal          | Total             |                           |                               |                                          |             |
| PuO <sub>2</sub> -Th <sup>232</sup> O <sub>2</sub> | 25/12.5/50/12.5             | 0.2695              | 0.674<br>(0.611)† | 1.394<br>(1.330)† | 13.7                      | -1.071                        | 0.290                                    | 0.403       |
| PuC-Th <sup>232</sup> C                            | 25/12.5/50/12.5             | 0.2380              | 0.731<br>(0.670)  | 1.480<br>(1.420)  | 11.6                      | -1.614<br>(+0.834)‡           | 0.294                                    | 0.339       |
| PuC-U <sup>238</sup> C                             | 20/10/70/0                  | 0.1957              | 0.823             | 1.704             |                           | -0.042<br>(+2.212)            | 0.381                                    | 0.365       |
| PuC-U <sup>238</sup> C                             | 22/11/67/0                  | 0.1805              | 0.886             | 1.719             | 7.7                       | +0.605                        | 0.386                                    | 0.344       |
| PuC-U <sup>238</sup> C*                            | 22/11/67/0                  | 0.185               | 0.863             | 1.622<br>(0.759)  | 8.8                       | -0.121                        | 0.386                                    | 0.338       |

\* Core: PuC-U<sup>238</sup>C; blanket: ThC.

† The values in parentheses are the values due to capture of neutrons by thorium only.

‡ Void effect due to total loss of sodium from the innermost region of core occupying one-third of the core volume.

\*\* Ex-pile inventory: 100%; load factor: 80%; output power: 1500 MW(th).



TABLE 4. CHARACTERISTICS OF THE SYSTEMS WITH DIFFERENT PLUTONIUM COMPOSITIONS

| Pu-fuel | Pu-composition (at. %) |                   |                   | Pu/U <sup>238</sup> | Breeding ratio |       | (y)<br>Doubling*<br>time | (%)<br>( $\Delta k$ ) <sub>rod</sub> <sup>†</sup> | (%)<br>( $\Delta k$ ) <sub>op</sub> <sup>‡</sup> | ( $\times 10^{-2}$ )<br>$\beta_{eff}$ | (μsec)<br>l |
|---------|------------------------|-------------------|-------------------|---------------------|----------------|-------|--------------------------|---------------------------------------------------|--------------------------------------------------|---------------------------------------|-------------|
|         | Pu <sup>239</sup>      | Pu <sup>240</sup> | Pu <sup>241</sup> | Pu <sup>242</sup>   | Internal       | Total |                          |                                                   |                                                  |                                       |             |
| Pu(A)   | 100                    | 0                 | 0                 | 0                   | 0.736          | 1.470 | 13.1                     | +0.180                                            | -1.049                                           |                                       |             |
| Pu(B)   | 74.7                   | 10.2              | 12.4              | 2.7                 | 0.836          | 1.641 | 8.7                      | -0.026                                            |                                                  | 0.357                                 | 0.338       |
| Pu(I)** | 77.4                   | 17.7              | 3.3               | 0.6                 | 0.831          | 1.628 | 9.3                      | +0.717                                            | -0.706                                           | 0.386                                 | 0.344       |
| Pu(G)   | 60                     | 25                | 12                | 3                   | 0.886          | 1.719 | 7.7                      | +0.605                                            |                                                  |                                       |             |

1. Fuel: PuC-U<sup>238</sup>C.2. Core: fuel/SS/Na; 22/11/67 vol. %. Blanket: U<sup>238</sup>C/SS/Na: 50/15/35 vol. %.

\* Ex-pile inventory: 100%; load factor: 80%; output power: 1500 MW(th).

† Reactivity change due to total loss of sodium from the core.

‡ Reactivity change due to  $7.5 \times 10^{-4}$  MWD of burnup.

\*\* Plutonium from Tokai reactor.

component of reactivity depends on the shape of the system. The spectra were calculated by the ESELEM. The effective cross-sections were collapsed into six groups and then two-dimensional calculations were carried out. The effect of sodium removal on the spectrum near low-lying resonance energies of sodium is shown in Fig. 3. It should be noted that the decrease of neutron population just below the resonance peak reduces the effective removal cross-section of every element. The sodium void effects calculated are listed in Table 1. The reactivity changes due to Doppler effect were also calculated by the code DOBROC, and the results are summarized in Table 2.

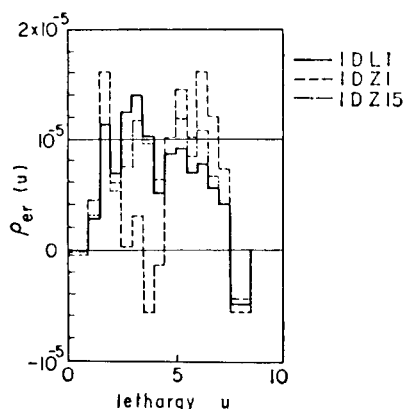


FIG. 4. Elastic removal components of sodium void effect.

The spectrum in the test zone deviates from that of the large core. As far as the total reactivity effect of sodium voids is concerned, this deviation does not have a large influence. The reactivity contributions from each group for the elastic removal component, however, were affected by this deviation. To eliminate this deviation, a buffer region is placed between the test and the driver zone. The effects are seen in Fig. 4.

The sodium-void reactivity  $(\Delta k)_{\text{sod}}$  due to total loss of sodium from the core, breeding properties and kinetic parameters of Pu-U and Pu-Th systems were studied.<sup>(11)</sup> The results are summarized in Table 3. The reactivity effect due to a partial loss of sodium from the active core depends mainly on the fuel volume fraction, and hence on the sodium volume fraction. The maximum fuel volume fraction, in which a positive reactivity effect is not caused by the total loss of sodium, is found to be about 5 vol. % larger for the Pu-Th system than for the Pu-U system.

The effects of fuel burnup upon the breeding ratio,  $k_{\text{eff}}$  and  $(\Delta k)_{\text{sod}}$  were investigated.<sup>(12)</sup> Only small variations were found in the breeding ratio. Thus, the initial breeding ratio may be a measure of the breeding ability of fast reactors through their life of operation. The effect of fuel isotopic com-

position was also investigated for different compositions of Pu, and the results are shown in Table 4.

To make the sodium-void reactivity more negative, it is effective to increase the ratio  $R_c$  of Pu atoms to U atoms by giving the system a larger leakage rate and/or smaller fuel volume fraction. However, a large value of  $R_c$  may yield a large reactivity loss  $(\Delta k)_{op}$  due to the fuel burnup. To control this opposite effect, a system, having an internal blanket loaded with Pu and U in the ratio of a few per cent, was studied. Because of the high leakage, the internal blanket system has a shorter effective neutron generation time than the conventional type fast reactors. The variation of  $(\Delta k)_{sod}$  due to  $3.0 \times 10^5$  MWd burnup amounts only to 0.125%. This effect does not seem to be serious.

### 5. Reactor Kinetics

The code TERESCOPE<sup>(13)</sup> has been developed to analyse the reactor behaviour before the boiling of sodium. This code is written for the IBM-7044 and consists of a combination of the following functions:

- (a) reactor kinetic equation;
- (b) reactivity coefficients, using six-group perturbation theory;
- (c) temperature distribution in the fuel and coolant, depending on their position in the reactor;
- (d) six-group neutron flux and adjoint flux, using Twenty Grand.

These computations, except (a), are done numerically for a two-dimensional cylindrical geometry. The combination of the above processes gives most of the quantities of interest for fast reactor kinetics. By using this code, it is expected to obtain precise knowledge in the pre-accident stage of the reactor.

The next interest is the method to measure the various power coefficients of reactivity. The study of reactor kinetics is usually carried out by measuring the reactor power-response for external reactivity disturbance to the reactor. It has been believed the sine oscillator is the best for ordinary purposes. However, it is rather difficult to install the oscillator, especially in fast reactors, and another difficulty is concerned with the non-linearity due to small reactivity disturbances by the sine oscillator. As another type of reactivity disturbance, it is quite natural to consider the ramp-type reactivity disturbance. A researcher of JAERI joined the group for reactor kinetics study in EBR-II, and the results of this study have been reported.<sup>(14)</sup>

Rod drop experiments, which involve dropping a special control rod of  $4 \times 10^{-4} \Delta k/k$  in a time of 200 msec, were carried out in EBR-II. The reactor power changes at zero power and at high power, induced by inserting the

special control rod, were recorded on a magnetic tape recorder and digitized. By applying the inverse kinetics,  $k_{ex}$  as a function of time at zero power and at high power were obtained from these digitized data. The difference between  $k_{ex}$  (zero power) and  $k_{ex}$  (high power) is the so-called feed back reactivity. Typical examples of  $k_{ex}$  are shown in Fig. 5. By analysing this feed back reactivity step by step, feed back parameters may be obtained. The measurements taken at different power runs at several power levels have given the significant results that the prompt feed back power coefficient

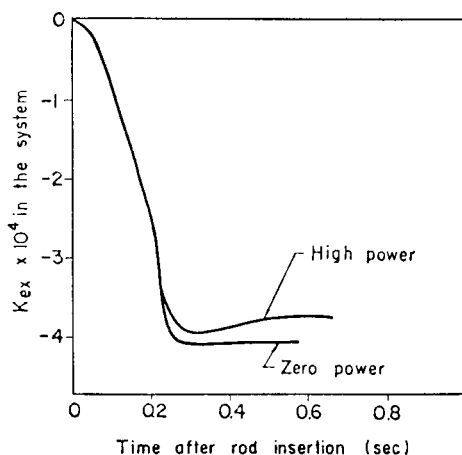


FIG. 5. Excess reactivity after rod insertion.

represents approximately 40% of the total feed back and has a time constant of about 0.3 sec. It has been reported that these values of feed back parameters reasonably agree with the theoretical values, and this kind of method appears useful in reactor kinetics studies in spite of its simplicity.

An attempt has also been made to derive a mathematical model by which the transfer function of fast reactors could be predicted accurately.<sup>(15)</sup> In this model, it is assumed for EBR-I Mark IV that the expansion and deformation of the core structure depend not only on the temperature gradient of the coolant in its subassembly but also on the thermal deformation of the upper structure. This assumption appears to hold good, when each fuel element consists, from top to bottom, of a plenum, upper blanket, core, and lower blanket, since the radial mixing of coolant can then be almost neglected. A detailed distributed-parameter model is introduced, in which the temperatures of the fuel and coolant are evaluated as continuous radial-axial and axial functions, respectively, assuming that there is no heat conduction in the longitudinal direction and that the axial power distribution is sinusoidal.

When the average coolant temperature from the core to the upper structure is used to estimate the core structure expansion, the calculated over-all transfer function agrees closely with the measured values for the gain and the phase, and especially for the peak frequency.

## 6. Experimental Techniques

As one of the preliminary studies on fast reactor physics, an exponential experiment was carried out.<sup>(16)</sup> The system consisted of 2.1 tons natural uranium, stainless steel, aluminium, and voids, and the neutron source was an enriched uranium converter plate attached to the surface of a thermal column of JRR-1. The primary purposes of this experiment were to study the fast neutron behaviour in natural uranium systems and to develop experimental techniques, especially of neutron spectrum measurement. The diffusion parameters, the fission ratio, the capture rate of  $U^{238}$ , and neutron spectra were measured.

In the experimental techniques, a method, based on the proton-recoil method using a nuclear emulsion, was developed<sup>(17)</sup> for obtaining the angular distribution and energy spectrum of an anisotropic fast neutron flux. This method involves measuring the proton-recoil spectra in two directions in the emulsion plate, exposed to the neutron flux within the system. The threshold reactions, namely,  $In^{115}(n, n') In^{115m}$ ,  $Ni^{58}(n, p) Co^{58}$ ,  $S^{32}(n, p) P^{32}$ ,  $Fe^{56}(n, p) Mn^{56}$ , and  $Al^{27}(n, \alpha) Na^{24}$  were used to obtain the neutron spectrum in the system. These threshold cross-sections have been examined, and programming for the spectrum analysis by a computer is now in progress.

The applications of semi-conductor counters for neutron measurements have been investigated,<sup>(18)</sup> and a  $Li^6$ -sandwiched neutron spectrometer is now prepared for spectrum measurements in the system. A proton-recoil fast neutron spectrometer was also developed.<sup>(19)</sup> This spectrometer consists of three methane-filled proportional counters, arranged axially with one common centre wire. An insulator disk with many holes is inserted between the first and the second counters, which acts as a proton collimator. Signals from the first and second cathodes are used as the coincidence signals, and those from the third cathode as the anti-coincidence signals. Signals from the centre wire give the neutron spectrum to be measured. The counter should be set in the same axis of the extracted neutron beam. The energy resolution depends on the collimator openings, and a higher resolution is obtained at the sacrifice of efficiency; an energy resolution of 13% was obtained for 0.6 MeV neutrons.

For noise analysis experiments, an energy- and space-dependent general theory<sup>(20, 21)</sup> was developed. This theory is expected to be very useful for the analysis of large cores, coupled systems, and subcritical systems, in which the space-dependent treatment is necessary.

### 7. Pulsed Neutron Experiments

Pulsed neutron experiments for non-multiplying and multiplying systems have been carried out,<sup>(22)</sup> using a 2 MeV Van de Graaff accelerator. The purpose of the experiments is to investigate the energy-dependent time behaviour of fast neutrons in the systems. The neutron burst of about 4 nano-second width, from the reactions of D-T and D-D, is generated with

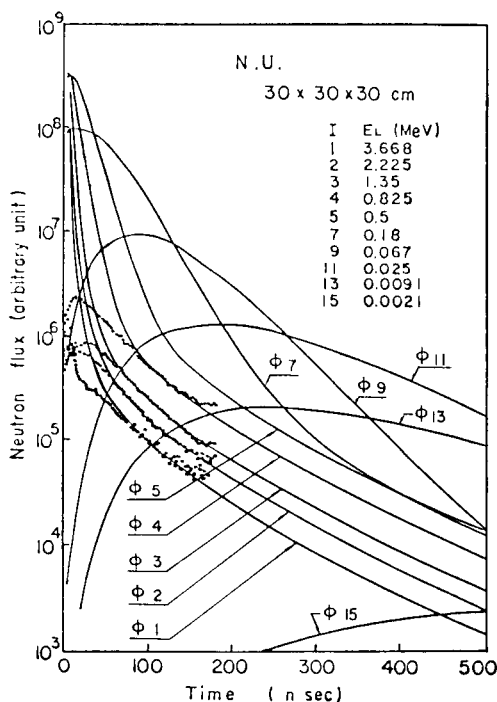


FIG. 6. Energy dependent fast neutron flux decay curves in a system of natural uranium.

the repetition of 200 nano-seconds at the centre of the cubic system. Leakage neutrons are detected on the surface with  $\text{Li}^6\text{I}$  scintillators. The time analyser consists of a Cranberg-type time-to-pulse height converter and a fast-slow coincidence circuit.

For lead systems, an exponential decay curve was observed for each energy range and the decay constant changed systematically with the energy. The experimental results for 20 cm<sup>3</sup> and 30 cm<sup>3</sup> natural uranium systems were compared with the theoretical curves, obtained by the code GS-1 based on the theory of Storrer and Stievenart. The results for the 30 cm<sup>3</sup> are shown in Fig. 6. Although the decay curves for higher energy neutrons show

the same pattern, it is obvious from the calculated curves that those for lower energy neutrons gradually rise and the persisting modes have not yet been formed as a whole. However, Fig. 6 shows some possibility of finding a certain constant for the system from the experimental results of the early part of the decay curves of the higher energy neutrons, where the detection is easier because of the abundance of neutrons and good energy resolution

## 8. Fast Critical Facility

The fast critical facility,<sup>(23)</sup> called the FCA, is now under construction as part of the fast breeder reactor development programme in Japan. This

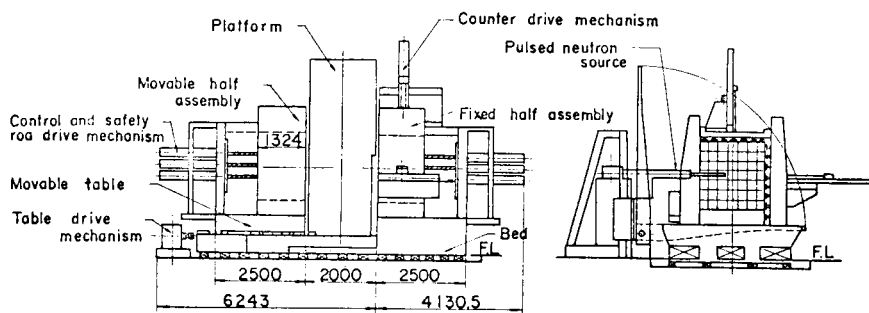


FIG. 7. Outline of FCA.

facility will be used in the investigation of the nuclear characteristics required for the design and evaluation of large fast reactors. The main design features of this facility are achievement of a high flexibility and of a sufficient reproducibility to permit high-accuracy measurements, with maximum of safety. To achieve these requirements, the split-table type of horizontal matrix structure was employed. Different core materials, in the form of small plates, are assembled in the matrix, so as to build any desired core configurations. The outline of the facility is shown in Fig. 7.

On each of the two tables are stacked horizontally  $5.52 \text{ cm}^2$  stainless steel tubes, constrained in position by side and upper frames, forming a half matrix. The matrix consists of 35 rows by 35 columns, but the facility is designed so as to be easily extended to 51 rows by 51 columns in which large cores up to 3000 l with a 50-cm thick blanket can be built. One table is supported by recirculating roller bearings and driven on the raceways of the bed by means of an electric motor, which turns the screw, engaged with a rack attached to the lower side of the table. As this table approaches and comes finally in contact with the other table, the speed of the table is changed in three stages. It may be mentioned here that the whole facility can withstand violent earthquakes of 0.6 g.

The core materials are mainly of the dimensions of  $5.08 \text{ cm}^2$ , with a thickness of 1.63 or 3.18 mm. The fuel used in the assembly will be 20% enriched uranium for the time being. Other materials used are natural uranium, canned sodium, aluminium, stainless steel, oxides of iron or aluminium and graphite. These plates are loaded in stainless steel drawers, which are then inserted into the matrix tubes. Each half of the assembly contains four or five safety-control dual purpose rods of fuel removal type. Usually, one of the rods is used as the control rod for each half, the rest being used as safety rods. On scram, all the rods are immediately driven out of the core by compressed air; the rods start to move within 35 msec after the neutron flux overrides the trip setting in the safety channels and travel within 200 msec over 80% of the full stroke 600 mm long.

Each half of the assembly is provided with an americium-beryllium neutron source of about five curies. The power level of the assembly is monitored by two fission chambers and six uncompensated  $\text{B}^{10}\text{F}_3$  ionization chambers. Two fission chambers are connected to pulse circuits and used as start-up channels. Two of the six ionization chambers are connected to the linear channels, the log N and period channels and the safety channels, respectively. This critical facility is also provided with a pulsed neutron source of a 200 keV Cockcroft-Walton-type accelerator which can generate neutron bursts from 0.05 to 10  $\mu\text{sec}$  width and from 1 to 200 kc repetition, with a maximum duty cycle of 5%, at the centre of the assembly.

The critical assembly is housed in a double container. The inner cell is a reinforced concrete structure of  $16 \text{ m}^3$  with the wall thickness of 11 m, and can withstand an internal explosion of 100 kg TNT with only small cracks. The outer cell is a reinforced concrete dome, 25 m in diameter and 21 m high, with the wall thickness of 0.4 m. The inner side is lined with 6 mm thick steel plate for leaktightness. This outer cell can withstand possible static pressure rises up to  $1.6 \text{ kg/cm}^2$  as a result of a serious accident, with the maximum leak rate being 2%/day. Near the double container there are four buildings, namely, a fuel handling building with a fuel vault, a control and laboratory building, an auxiliary building, and an accelerator building, in which another 300 keV pulsed neutron source is housed for the pulsed experiments of nano-second order. Two automatic fuel transport tubes connect the inner cell and the fuel handling building.

The facility will go critical at the beginning of 1967, with  $\text{U}^{235}$  fuel. During the first stage of experiments, the systems to be investigated are small fast systems or fast thermal coupled systems, owing to the insufficient amount of fuel available. The pulsed neutron source mentioned will be used for measuring the neutron energy spectra by the time-of-flight method, and for obtaining information concerning the kinetic behaviour of such systems, especially in the former systems. In the latter systems, the fast core of about 50 cm diameter will be surrounded by a buffer region and a thermal driver



region. These cores provide neutron spectra similar to those of large fast reactors, and it will be possible to compare the theoretical and experimental results for various parameters and to obtain information on the sodium void and Doppler effects. In the second stage of experiments, with the increased amount of  $U^{235}$  fuel and the Pu fuel available, these fast thermal coupled systems will be replaced by zoned systems to make the measurements of reactivity coefficients more accurate.

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